HYSEIS, or Hybrid Seismology Intelligence System, combines the Omori–Utsu and Gutenberg–Richter seismic models with XGBoost machine learning to forecast aftershock probabilities and magnitudes using historical earthquake data. It integrates fault-line mapping, distance estimation, ground-motion simulation, GIS-based hazard maps, and interactive 3D visualization to support seismic-risk analysis and disaster preparedness.